RESEARCH · 01

# *Should the Fouled Player Take the Penalty Himself?*

A ONE-TAILED Z-TEST · TOP 5 EUROPEAN LEAGUES

2026-05

renball

TL;DR

# Three sentences

THE QUESTION

*When a goalkeeper fouls a player who then takes the penalty himself, does conversion drop?*

THE TEST

*One-tailed two-proportion z-test. Treatment $n = 100$ (75.0%). Control $n = 3{,}757$ (78.25%).*

THE RESULT

*$\Delta = -3.25$ pp (predicted direction) but $p = 0.218$, one-tailed — fail to reject $H_0$ at the available power.*

THE THEORY

# *Why we'd expect a drop*

*The just-fouled player walks straight to the spot.*

Theory of the taker: physically jarred — body contact, hit the ground, adrenaline. Emotionally agitated, narrowed focus.

Theory of the keeper: locked in. The foul that gave the penalty was his fault. Nothing left to lose. Maximum motivation.

*→ If both pull conversion down, the bench should hand the ball to a designated specialist.*

DATA

# *What we built the test on*

| 4,110 | 3,322 | 100 | 5 |
| --- | --- | --- | --- |
| PENALTIES (LOOSE FILTER) | SINGLE-PENALTY MATCHES | TREATMENT-CELL N | LEAGUES |

Seasons in scope: 2017-2018 → 2023-2024

Competitions: Premier League, La Liga, Serie A, Bundesliga, Ligue 1.

Source: FBref player-match data, consolidated locally as FULL_DICT_ROWS.pkl.

Strict filter = matches with at most one penalty event (unambiguous GK-caused / taker-was-fouled attribution).

METHOD

# *One-tailed two-proportion z-test*

TREATMENT

## *GK fouled the taker*

Matches where the goalkeeper conceded the foul AND the fouled player is the same player who took the penalty.

CONTROL

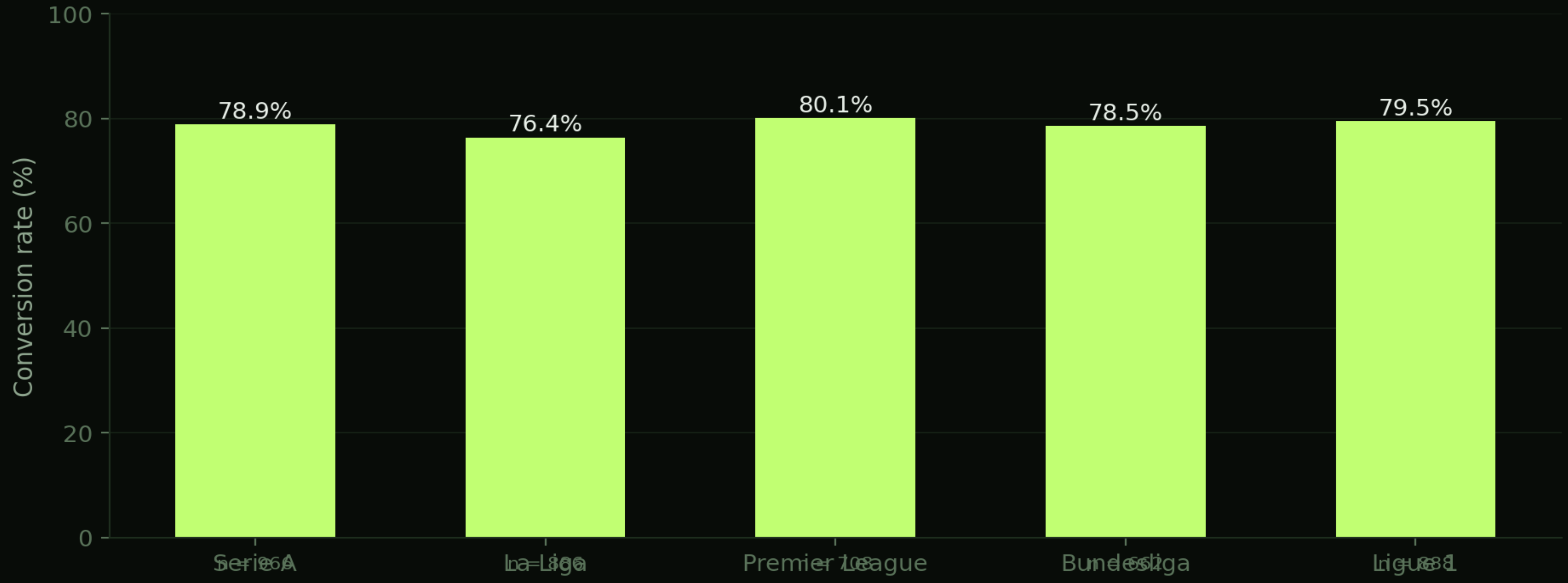
## *All other penalties*

Every other strict-filter match — GK fouled but a different player took, or outfielder caused the foul (any taker).

$H_0$: conversion_treatment = conversion_control · $H_1$: conversion_treatment < conversion_control (one-tailed, α = 0.05) · 95% Wilson CIs · statsmodels.proportions_ztest

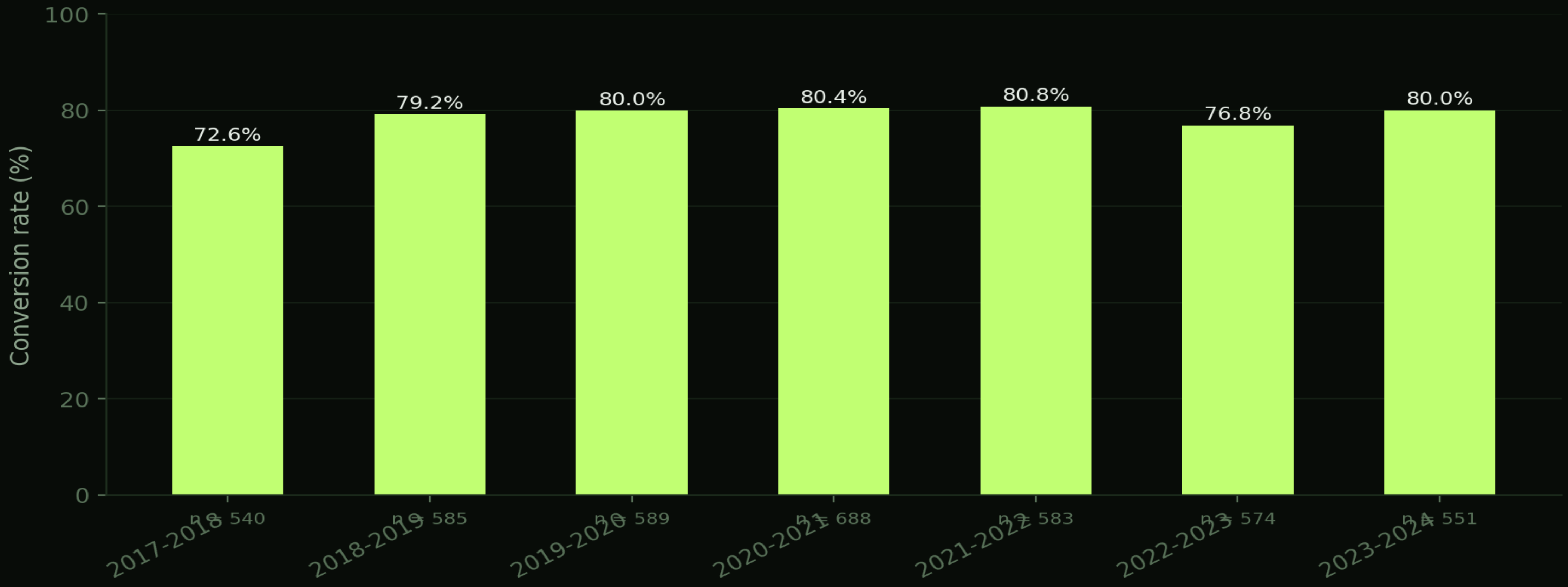
CONTEXT

# *Conversion rate by league*




Across the top 5 leagues conversion clusters tightly in the high 70s — league of origin is not a major source of variation.

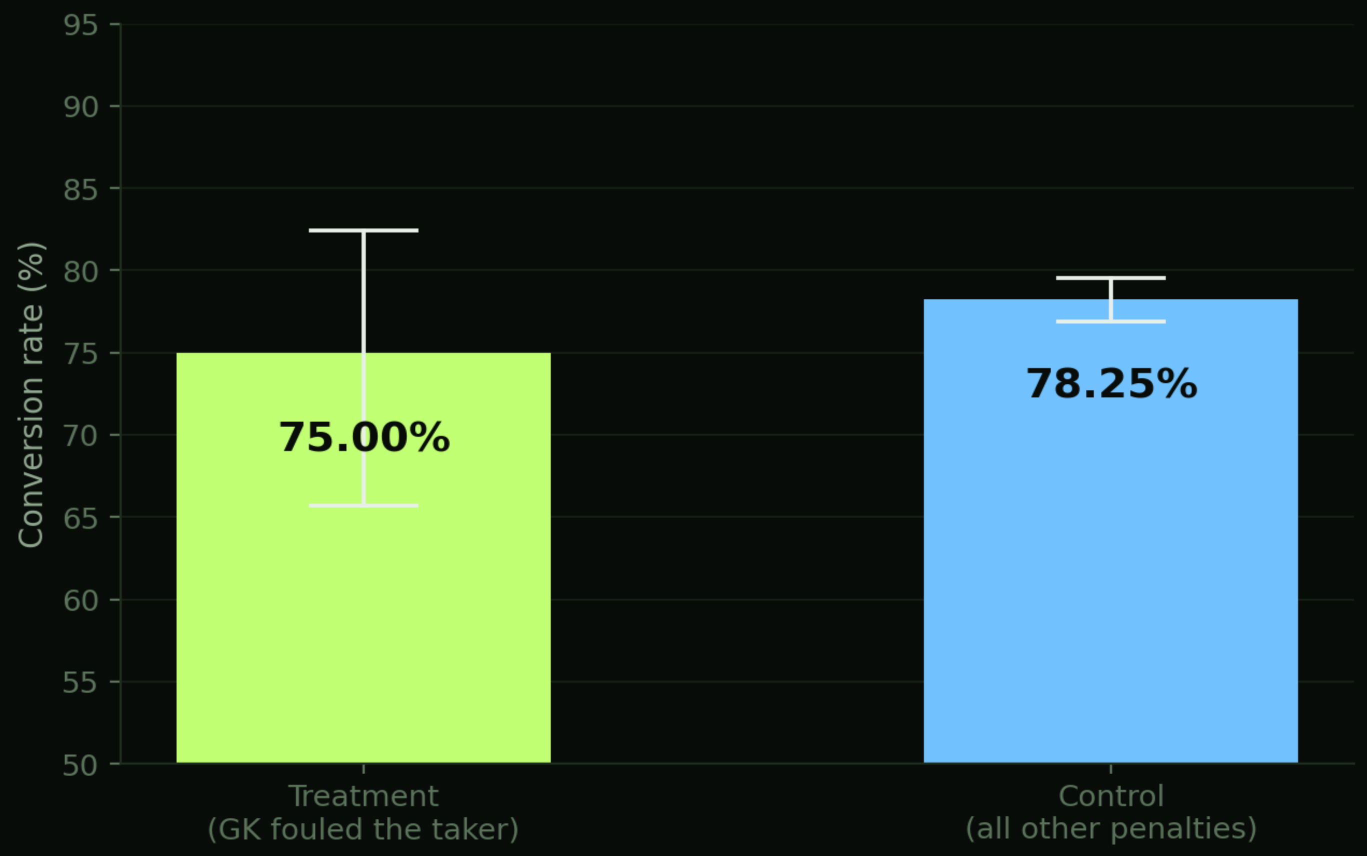
# *Conversion across seasons*




Year-on-year stability is high — the ~77-80% band holds across the period in scope.

PRIMARY TEST · ONE-TAILED

# *Treatment vs control*




TREATMENT

*75.00%*

n = 100 · 95% CI [65.70%, 82.45%]

CONTROL

*78.25%*

n = 3,757 · 95% CI [76.91%, 79.54%]

$\Delta$ = -3.25 pp   z = -0.777

p = 0.2185 (one-tailed) · Fail to reject $H_0$

Whiskers: 95% Wilson confidence intervals. Treatment CI is wide — n is the binding constraint.

# Where the *treatment cell* sits

| | GOALKEEPER CAUSED THE FOUL | OUTFIELDER CAUSED THE FOUL |
|---|---|---|
| TAKER = THE PLAYER WHO WAS FOULED | TREATMENT<br>*75.00%*<br>N = 100 | *78.87%*<br>N = 549 |
| TAKER = A DIFFERENT PLAYER | *78.11%*<br>N = 265 | *78.15%*<br>N = 2,943 |

Accent-bordered cell = treatment. The other three pool into control.

PER-LEAGUE REPLICATION

# The same test, *league by league*

| LEAGUE | T. RATE | T. N | C. RATE | C. N | Z | P (1-T) | SIG |
|---|---|---|---|---|---|---|---|
| Serie A | 80.95% | 21 | 77.70% | 888 | +0.354 | 0.6383 | No · low n |
| La Liga | 57.89% | 19 | 77.46% | 803 | -2.003 | 0.0226 | Yes · low n |
| Premier League | 88.00% | 25 | 78.59% | 626 | +1.131 | 0.8710 | No · low n |
| Bundesliga | 61.54% | 13 | 78.18% | 614 | -1.429 | 0.0765 | No · low n |
| Ligue 1 | 77.27% | 22 | 79.42% | 826 | -0.245 | 0.4030 | No · low n |

Per-league treatment cells of 13–25 leave each test severely underpowered — pooled is the only inference to take seriously.

CONCLUSION

# What the data *says*

## The bench answer: *can't yet tell.*

Conversion is 3.25 pp lower when the fouled player takes the kick himself (75.00% vs 78.25% for everyone else). The point estimate moves in the predicted direction, but the treatment cell is too small to rule chance out ($p = 0.218$, one-tailed).

*→ No statistical warrant to override the fouled player who wants to take it himself. Equally, no clean evidence he's at full strength.*

# *Full methodology, source code and data:*

Full methodology, source code and data:

*renball.com*

/PROJECTS/01-PENALTY-ANALYSIS/

→ methodology formal $H_0$ / $H_1$, test assumptions, limitations

→ analysis.html interactive dashboard, live numbers from data.json

→ code/build.py Python pipeline — raw pickles → aggregated JSON